

How Does the Retinal Rod Na-Ca+K Exchanger Regulate Cytosolic Free Ca^{2+} ?

(Received for publication, January 18, 1995, and in revised form, March 30, 1995)

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The roles of 1) inactivation of Na-Ca+K exchange and 2) Ca^{2+} release from discs in regulation of cytosolic free Ca^{2+} were examined in intact rod outer segments (ROS) purified from bovine retinas. Measurements of cytosolic free Ca^{2+} (with fluo-3) were combined with Ca^{2+} flux measurements (^{45}Ca) in ROS that contained about 600 μM total Ca^{2+} . Na^{+} -induced Ca^{2+} extrusion was measured in a Ca^{2+} -free medium and did not lower cytosolic free Ca^{2+} to below 1 nM as expected from a coupling stoichiometry of $4\text{Na}^{+}:(1\text{Ca}^{2+}+1\text{K}^{+})$. Instead, cytosolic free Ca^{2+} was rapidly (20 s) lowered from about 1300 nM to 100–150 nM, while at the same time about 35% of total ROS Ca^{2+} was removed. During the next 40 min cytosolic free Ca^{2+} remained virtually steady, but total ROS Ca^{2+} was reduced by a further 50% at a 100-fold lower rate than that observed for the initial fast phase. The steady cytosolic Ca^{2+} concentration resulted from Ca^{2+} release from discs and subsequent removal across the plasma membrane by Na-Ca+K exchange operating at a greatly reduced rate. Addition of the alkali cation channel ionophore gramicidin led to a persistent increase in cytosolic free Ca^{2+} concentration to about 400 nM, presumably caused by an increase in intracellular Na^{+} . It is suggested that cytosolic free Ca^{2+} is not determined by the $\text{Na}^{+}:\text{Ca}^{2+}$ coupling ratio of the exchanger, but rather by a sensor on its cytoplasmic domain that controls inactivation of the Ca^{2+} extrusion mode and is sensitive to intracellular Ca^{2+} , Na^{+} , and K^{+} .

Ca^{2+} homeostasis in retinal rod outer segments (ROS)¹ is thought to be dominated by dynamic fluxes across the plasma membrane: Ca^{2+} influx is mediated by the light-sensitive cGMP-gated channels, and Ca^{2+} efflux is carried out by the Na-Ca+K exchanger (1). In darkness, a kinetic equilibrium between the above two Ca^{2+} fluxes keeps cytosolic relatively high. Light causes a decrease in Ca^{2+} influx, while continued Ca^{2+} efflux via the exchanger lowers cytosolic Ca^{2+} , and this is thought to mediate, at least in part, the process of light adaptation in both retinal rods and cones (2, 3). The coupling stoichiometry of the ROS Na-Ca+K exchanger has been shown to

be $4\text{Na}^{+}:(1\text{Ca}^{2+}+1\text{K}^{+})$ (4, 5), from which, under physiological conditions, a thermodynamic equilibrium free Ca^{2+} concentration of below 1 nM can be calculated (6). Therefore, the Na-Ca+K exchanger is expected to lower cytosolic free Ca^{2+} below 1 nM when Ca^{2+} influx via the cGMP-gated channels is interrupted for a prolonged time, e.g. when rods are saturated under bright daylight illumination.

Most of our knowledge on Na-Ca or Na-Ca+K exchangers is derived from studies of macroscopic Ca^{2+} fluxes or exchange currents when these exchangers operate far from thermodynamic equilibrium. In the absence of cGMP and cGMP-gated Ca^{2+} fluxes, the Na-Ca+K exchanger appears to be the only cation transporter operating in the ROS plasma membrane (7–9). In three previous studies we have loaded the cytosol of purified bovine ROS with the fluorescent Ca^{2+} -indicating dye fluo-3 and combined measurements of changes in cytosolic free Ca^{2+} with ^{45}Ca flux measurements (10–12). Unexpectedly, it proved difficult to lower free cytosolic Ca^{2+} below 50–100 nM upon stimulation of the Ca^{2+} efflux mode of the exchanger, even if thermodynamic considerations would predict a free Ca^{2+} concentration of less than 1 nM, and despite the ability of the Ca^{2+} influx mode of the exchanger to operate at very low external free Ca^{2+} concentrations of 10–20 nM. We concluded that the Ca^{2+} extrusion mode of the exchanger operates in a brief burst (10–30 s) of high velocity activity followed by a nearly complete inactivation, a novel feature for Na-Ca(+K) exchangers. As this may present a mechanism to prevent lowering of cytosolic free Ca^{2+} to undesirably low values when rods are saturated with light for prolonged periods of time, this study examines 1) the activity of the plasma membrane Na-Ca+K exchanger on time scales between a few seconds and 40 min, 2) the contribution of internal Ca^{2+} stores to Ca^{2+} homeostasis under those conditions, and 3) factors that determine at which free Ca^{2+} concentration the Ca^{2+} efflux mode of the exchanger inactivates. The significance of our findings for the mechanism(s) employed by ROS to set intracellular free Ca^{2+} is discussed.

MATERIALS AND METHODS

All experimental procedures employed in this study have been described in detail elsewhere and are only briefly outlined here.

ROS Preparation—Bovine retinas were dissected from fresh eyeballs obtained at a local abattoir and kept in a light-tight container. Intact Ca^{2+} -depleted bovine ROS were purified on mixed sucrose-Ficoll gradients containing Na^{+} and EDTA and washed with sucrose-EDTA solutions (13). ROS were stored in concentrated suspensions (200–300 μM) at 4 °C in 600 mM sucrose, 5% w/w Ficoll 400, 20 mM Hepes (pH 7.4), and 0.1 mM EDTA (pH in this and all other solutions was adjusted to 7.4 with arginine). Up to this point, all procedures were carried out under dim red illumination.

Fluorescence Measurements of Free Cytosolic Ca^{2+} —A suspension of bleached intact and Ca^{2+} -depleted bovine ROS (final opsin concentration 100 μM) was incubated for 45 min at room temperature with 50 μM fluo-3AM in a medium containing 400 mM sucrose, 20 mM Hepes (pH 7.4), 10 mM KCl, 1 mM HEDTA, 0.25 mM CaHEDTA, and ^{45}Ca (for the

* This research was funded through an operating grant from the Canadian Medical Research Council. The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

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¹ The abbreviations used are: ROS, rod outer segment; HEDTA, N-hydroxyethylethylenediaminetriacetic acid; FCCP, carbonyl cyanide p-trifluoromethoxyphenylhydrazone.

flux experiments only). The unhydrolyzed fluo ester and unincorporated ^{45}Ca was removed by sedimentation of ROS through a cushion of 600 mM sucrose, 20 mM Hepes (pH 7.4). Further experimental procedures and calibration procedures were carried out as described before (10, 12). Fluorescence measurements were carried out in an SLM-Aminco series 2 luminescence spectrometer (SLM Instruments, Urbana, IL). ^{45}Ca fluxes in ROS were measured with a rapid filtration technique over borosilicate glass fiber filters as previously described (8). Calculations of total Ca^{2+} fluxes assume that bovine ROS contain an overall rhodopsin concentration of 3 mM or 2.9×10^7 rhodopsin molecules.

RESULTS

Accessibility of ROS Ca^{2+} to Removal by Na-Ca or Sr-Ca Exchange—The objective of the experiments reported in this study was to analyze compartmentalization of Ca^{2+} fluxes in isolated intact bovine ROS and its relationship to the observed inability of the Ca^{2+} extrusion mode of the ROS Na-Ca+K exchanger to lower free cytosolic Ca^{2+} to the low values expected from the coupling stoichiometry of $4\text{Na}^+:(1\text{Ca}+1\text{K})$ (11, 12, 14). Analysis of both total internal Ca^{2+} , cytosolic free Ca^{2+} and internal Ca^{2+} sequestration with sufficient analytical precision necessitated to prepare first ROS that contained no measurable Ca^{2+} (Na^+ -enriched and Ca^{2+} -depleted ROS were prepared by isolating and purifying ROS in Ca^{2+} -free and Na^+ -containing media). Subsequently, Ca^{2+} -depleted ROS were loaded with Ca^{2+} , ^{45}Ca and fluo-3, and stored in a Na^+ - and Ca^{2+} -free medium; the Ca^{2+} extrusion mode of the Na-Ca+K exchanger was initiated by addition of Na^+ . A Ca^{2+} -free medium was used to eliminate Ca^{2+} influx from the external medium. The above experimental paradigm is nonphysiological, but it enables us to analyze compartmentalization of Ca^{2+} and Ca^{2+} fluxes without ambiguity. In the first two figures experiments are illustrated that were designed to establish localization of total ROS Ca^{2+} to the cytosol and to internal stores, respectively. Ca^{2+} -depleted ROS were loaded with Ca^{2+} (at an external free Ca^{2+} concentration of 350 nM) and ^{45}Ca to yield moderate Ca^{2+} loads of 0.1–0.3 mol Ca^{2+} /mol rhodopsin, equivalent to a total ROS Ca^{2+} content of 300–900 μM . Ca^{2+} -loaded ROS were stored in a Na^+ -free buffered sucrose solution and both cytosolic free Ca^{2+} and total Ca^{2+} levels were stable for 24 h when ROS were stored as a concentrated suspension (200–300 μM rhodopsin) at 4 °C. Different aliquots of the same ROS suspension were diluted to a final rhodopsin concentration of 1–2 μM , NaCl was added to the external medium to initiate Ca^{2+} extrusion via the Na-Ca+K exchanger and either fluo-3 fluorescence was measured to monitor cytosolic free Ca^{2+} or ^{45}Ca radioactivity was measured to monitor total ROS Ca^{2+} . Addition of NaCl caused a rapid Ca^{2+} efflux phase lasting about 20 s, during which most of the cytosolic Ca^{2+} was removed from ROS (free Ca^{2+} dropped from 1300 nM to 150 nM), and during which about 40% of total ROS Ca^{2+} was removed (Fig. 1, left panels). The ROS Na-Ca+K exchanger also executes Ca-Ca self-exchange fluxes, in which Sr^{2+} can replace Ca^{2+} (8, 15). Sr^{2+} can also replace Ca^{2+} in binding to fluo-3, but at a reduced fluorescence yield. When 0.5 mM SrCl_2 was added to Ca^{2+} -loaded ROS, a fast Sr-Ca exchange phase replaced cytosolic Ca^{2+} by Sr^{2+} and reduced the fluo-3 fluorescence (Fig. 1). The half-time for both Na^+ - and Sr^{2+} -induced lowering of Ca^{2+} fluo-3 fluorescence in the ROS cytosol was 4–7 s. Consistent with a rapid removal of cytosolic Ca^{2+} via both Sr-Ca and Na-Ca exchange, both Sr^{2+} and Na^+ caused a rapid release of about 40% of the total Ca^{2+} present in ROS. Similar experiments in over 50 Ca^{2+} -loaded ROS preparations with total Ca^{2+} loads ranging from 300 μM to 4 mM showed that consistently 40–50% of total internal Ca^{2+} could be rapidly exchanged against either Sr^{2+} or Na^+ and was probably localized to the cytosol.

Subsequent to the first 20 s of rapid Ca^{2+} release, Sr-Ca

exchange caused little further change in fluorescence during the next 20 min, while Na-Ca exchange gradually lowered cytosolic Ca^{2+} from a value of 150 nM to 100 nM after 20 min (Fig. 1, bottom right-hand panel). Therefore, the Na-Ca+K exchanger mediated very little net Ca^{2+} removal from the cytosol in the time interval between 20 s and 20 min after addition of Sr^{2+} and Na^+ , respectively. In contrast, slow components were observed for total Ca^{2+} efflux from ROS (measured with ^{45}Ca , Fig. 1, top right-hand panel); both Na-Ca and Sr-Ca exchange removed about 35% of the initial ROS Ca^{2+} content (^{45}Ca at time 0) in the period between 20 s and 10 min after addition of Na^+ and Sr^{2+} , respectively, whereas 1 h after addition of Na^+ or Sr^{2+} still 10–15% of the initial ^{45}Ca content remained (not illustrated).

Localization of Slowly Exchanging Ca^{2+} to an Internal Membranous Store—The experiment in Fig. 1 illustrates that the slow component of both Na-Ca exchange and Sr-Ca exchange could change the total ^{45}Ca content of ROS without a concomitant change in cytoplasmic Ca^{2+} , suggesting that this slow component represents Ca^{2+} removal from some internal compartment, i.e. the intradiscal space. Two approaches were used to substantiate the above conclusion, one employing selective permeabilization of disc membranes by A23187, and the second by selective permeabilization of the plasma membrane with streptolysin S.

Low doses of A23187 (<200 nM) selectively permeabilize the disc membranes and release intradiscal Ca^{2+} into the cytosol resulting in a rise in cytosolic free Ca^{2+} (12). Here, the amplitude of the A23187-induced increase in cytosolic Ca^{2+} was used as an indicator of intradiscal Ca^{2+} content as illustrated in Fig. 2. Superimposed fluo-3 fluorescence traces are shown of Sr-Ca exchange; these traces were obtained with a series of separate cuvettes to which 100 nM A23187 was added at different time points (as indicated by the upward pointing arrows) after initiation of Sr-Ca exchange at time 0 (just prior to addition of A23187, 2 mM EDTA was added to chelate all external Sr^{2+} and Ca^{2+} and stop Sr-Ca exchange across the plasma membrane). Initially, addition of A23187 caused a large increase in fluorescence to indicate Ca^{2+} release from internal stores that competed with cytosolic Sr^{2+} for binding to fluo-3. The amplitude of the A23187-induced rise in fluorescence gradually decreased as the interval between Sr^{2+} addition and addition of EDTA/A23187 was increased indicating progressive loss of Ca^{2+} from the internal store. The time course with which the amplitude of the A23187-induced fluorescence signals decreased was quite similar to the time course of the slow phase of Sr^{2+} -induced ^{45}Ca efflux (Fig. 2, inset), and both are consistent with removal of Ca^{2+} originally localized within a membranous internal compartment, i.e. the intradiscal space.

Further support for the above localization of internal Ca^{2+} to cytosolic and intradiscal pools, respectively, was obtained by selective permeabilization of the ROS plasma membrane with streptolysin S. Streptolysin S caused the complete release of fluo-3 from ROS, whereas only a partial release of ^{45}Ca was observed that matched the amount released by the rapid component of Na-Ca exchange (not illustrated).

Internal Ca^{2+} Release from ROS Discs—The experiment illustrated in Fig. 1 implies that internal Ca^{2+} stores in isolated ROS lost Ca^{2+} after the initial rapid component of Na-Ca or Sr-Ca exchange had removed most cytosolic Ca^{2+} . To demonstrate Ca^{2+} release from discs into the cytosol, Na^+ -stimulated Ca^{2+} release was allowed to proceed for 30 s (which extruded most cytosolic Ca^{2+}) and then stopped by a 40-fold dilution into a Na^+ -free medium (dilution medium contained 5 mM Mg^{2+} to inhibit residual Na-Ca exchange at 1 mM external Na^+ and 1 mM EGTA to prevent Ca^{2+} uptake from the external medium).

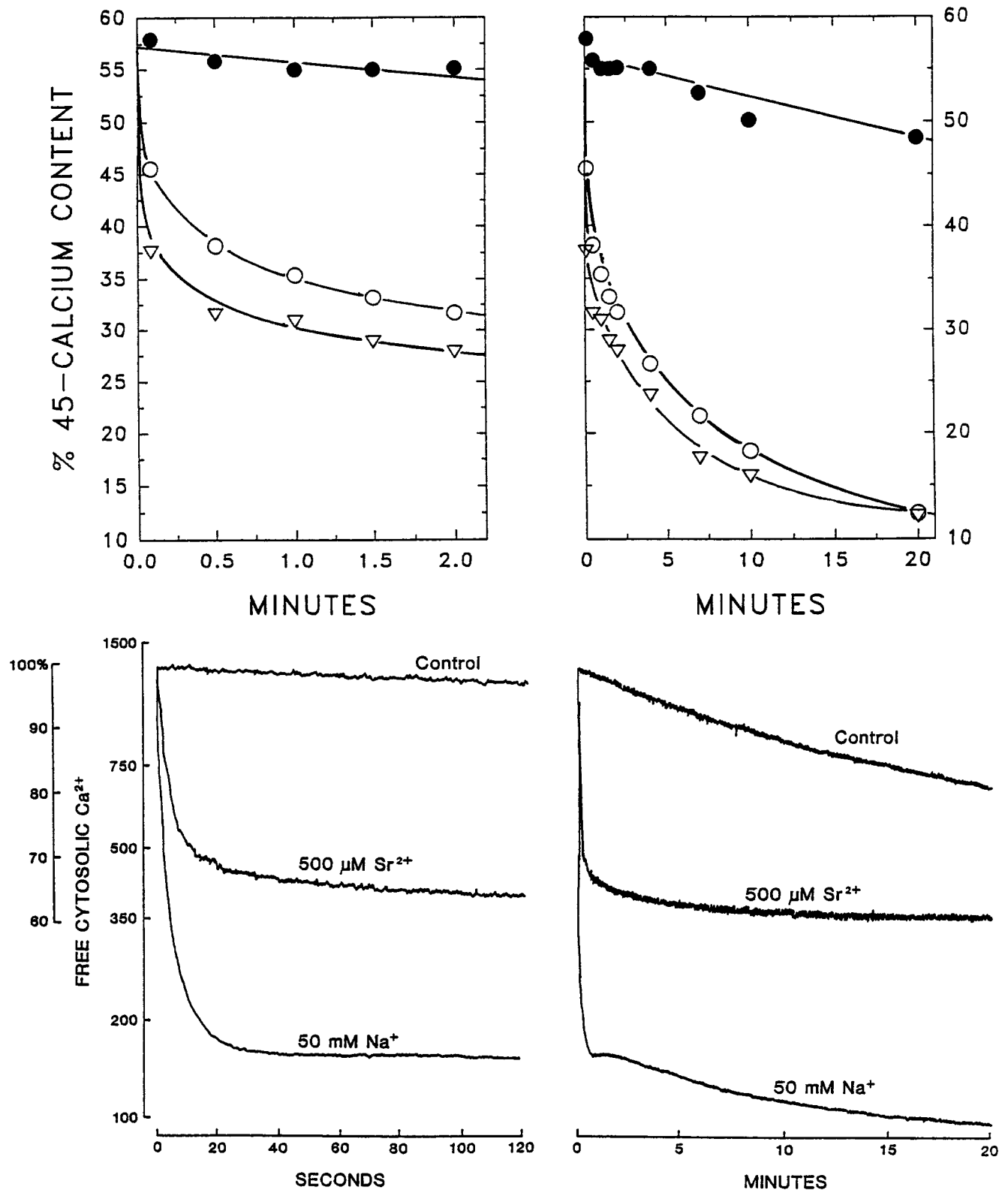


FIG. 1. Comparison between changes in cytosolic and total Ca^{2+} in bovine ROS induced by Na-Ca and Sr-Ca exchange. Ca^{2+} -, ^{45}Ca -, and fluo-loaded intact bovine ROS were prepared as described under "Materials and Methods" and were incubated in 600 mM sucrose, 20 mM Hepes (pH 7.4), 2 mM KCl, 500 nM FCCP, 0.1 mM EDTA at a final opsin concentration of 1.5 μM . Sr^{2+} - or Na^{+} -stimulated Ca^{2+} release was initiated at time 0 by addition of 500 μM SrCl_2 (open circles) or 50 mM NaCl (inverted triangles); controls represent changes in ^{45}Ca content (filled circles in top panels) or in free cytosolic Ca^{2+} (bottom panels) without additions to the incubation medium. Incorporation of ^{45}Ca in ROS in this and subsequent figures is expressed as the percentage of total ^{45}Ca present. Temperature, 25 °C.

When Ca^{2+} removal across the plasma membrane via the Na-Ca+K exchanger was minimized this way, a large rise in cytosolic free Ca^{2+} from about 150 nM to about 600 nM was ob-

served, whereas in the continued presence of 50 mM Na^{+} a further drop to about 80 nM was observed (Fig. 3, left-hand panel). Measurements of the ^{45}Ca content showed that dilution

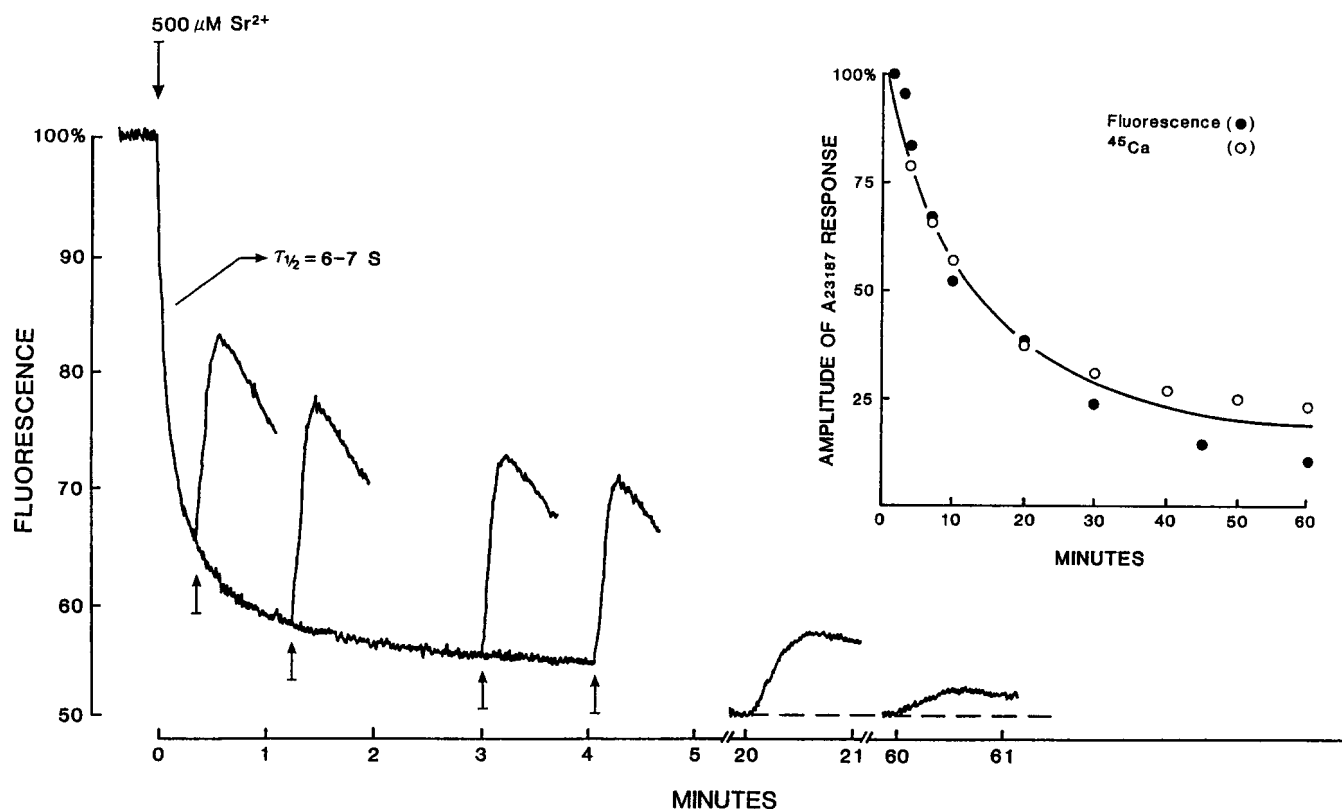


FIG. 2. **Sr-Ca exchange removes both cytosolic Ca^{2+} and intradiscal Ca^{2+} .** Experimental conditions were as described in the legend of Fig. 1. Seven superimposed traces are illustrated that represent seven different cuvettes. In all cases Sr-Ca exchange was initiated at time 0 by addition of 500 μM SrCl_2 . The upward pointing arrows (as well as at 20 and 60 min) indicate addition of 2 mM EDTA to chelate external Sr^{2+} and addition of 100 nM A23187 to release internally sequestered Ca^{2+} . The insert in the top right corner compares the amplitude of the A23187 signals (filled circles) with the remaining ^{45}Ca content of ROS (open circles); in the latter case, 100% represents the ^{45}Ca content after the fast Sr-Ca exchange phase had removed cytosolic ^{45}Ca . Temperature, 25 $^{\circ}\text{C}$.

into the Na^+ -free, Mg^{2+} -containing medium greatly reduced net Ca^{2+} loss from ROS compared with the loss observed in the continued presence of 50 mM Na^+ (Fig. 3, *right-hand panel*, compare *filled triangles* with *open squares*). Free cytosolic Ca^{2+} was lowered during the first 5 s following addition of 50 mM Na^+ at an average rate of 200 nM/s. In contrast, the subsequent slow phase of Na^+ -induced Ca^{2+} release took 10 min to lower free cytosolic Ca^{2+} from 150 nM to 100 nM at an average net rate of Ca^{2+} efflux of 0.08 nM/s. The initial rate of Ca^{2+} release from discs was between 2 and 4 nM/s (as judged from the dilution experiment) and it took 10 min to raise cytosolic free Ca^{2+} from 150 nM to 500 nM (average rate over 10 min of 0.6 nM/s).

Fig. 3 also illustrates two cases in which Ca^{2+} release from the discs could occur without first lowering cytosolic free Ca^{2+} to <200 nM. First, ^{45}Ca release in Na^+ - and Ca^{2+} -free medium was greatly reduced by addition of 5 mM Mg^{2+} to the external medium (compare *filled triangles* with *filled circles*), perhaps by stabilizing the ROS structure. A steady rise in cytosolic free Ca^{2+} was observed in the fluo-3 traces under this condition indicating Ca^{2+} release from discs. Second, Ca^{2+} release was measured at a low external Na^+ concentration of 1 mM resulting in a modest Ca^{2+} extrusion rate due to low occupancy of the Na^+ binding sites (Na^+ dissociation constant is 35 mM) (13, 16). At 1 mM Na^+ , a concurrent Ca^{2+} release was observed in cytosolic free Ca^{2+} measurements and total ^{45}Ca measurements, *i.e.* no separation in time existed between net Ca^{2+} efflux from the cytosol and net intradiscal Ca^{2+} loss as was observed at high Na^+ .

Intracellular Cation Dependence of Cytosolic Free Ca^{2+} Set by the Na-Ca+K Exchanger—Gramicidin is an alkali cation-selective channel ionophore that greatly facilitates Na^+ and K^+

entry across the ROS plasma membrane (16). Gramicidin caused a large rise in cytosolic Ca^{2+} (from 153 nM to a maximal value of about 450 nM) when added 15 s after initiation of Na^+ -induced Ca^{2+} efflux (Fig. 4, *left panel*). In contrast, the rate of net ^{45}Ca efflux from ROS was little affected by addition of gramicidin and indicated a continued net efflux of Ca^{2+} across the plasma membrane (Fig. 4, *right panel*). While the ^{45}Ca efflux rate gradually declined in the 40 min following Na^+ addition, the gramicidin-induced rise in Ca^{2+} persisted. The gramicidin-induced rise in cytosolic free Ca^{2+} was dependent on Ca^{2+} release from discs, since it was greatly reduced by pretreatment of ROS with 100 nM A23187 (not illustrated). Pretreatment with 100 nM A23187 caused selective Ca^{2+} permeabilization of the disc membrane and enabled subsequent Na^+ -dependent Ca^{2+} extrusion to remove intradiscal Ca^{2+} released by A23187 before inactivation occurred (12).

High external K^+ did not prevent the gramicidin-induced and Na^+ -dependent rise in Ca^{2+} *per se* (not shown), but it abolished the persistent nature of the gramicidin-induced rise in Ca^{2+} . When K^+ in the medium was raised from 5 to 55 mM at the peak of the gramicidin-induced rise in Ca^{2+} , cytosolic free Ca^{2+} was lowered to a new steady value of about 100 nM, while ^{45}Ca efflux across the plasma membrane was slightly increased (Fig. 4). The above experimental protocol was repeated by increasing the external K^+ concentration to different values, and this resulted in different cytosolic free Ca^{2+} concentrations (Fig. 5; Li^+ could not replace K^+). The data illustrated in Figs. 3 and 4 were obtained with the same ROS preparation, but very similar results were obtained in at least six other experiments with different ROS preparations.

From the above results it might be anticipated that the gramicidin-induced rise in cytosolic Ca^{2+} would mirror that

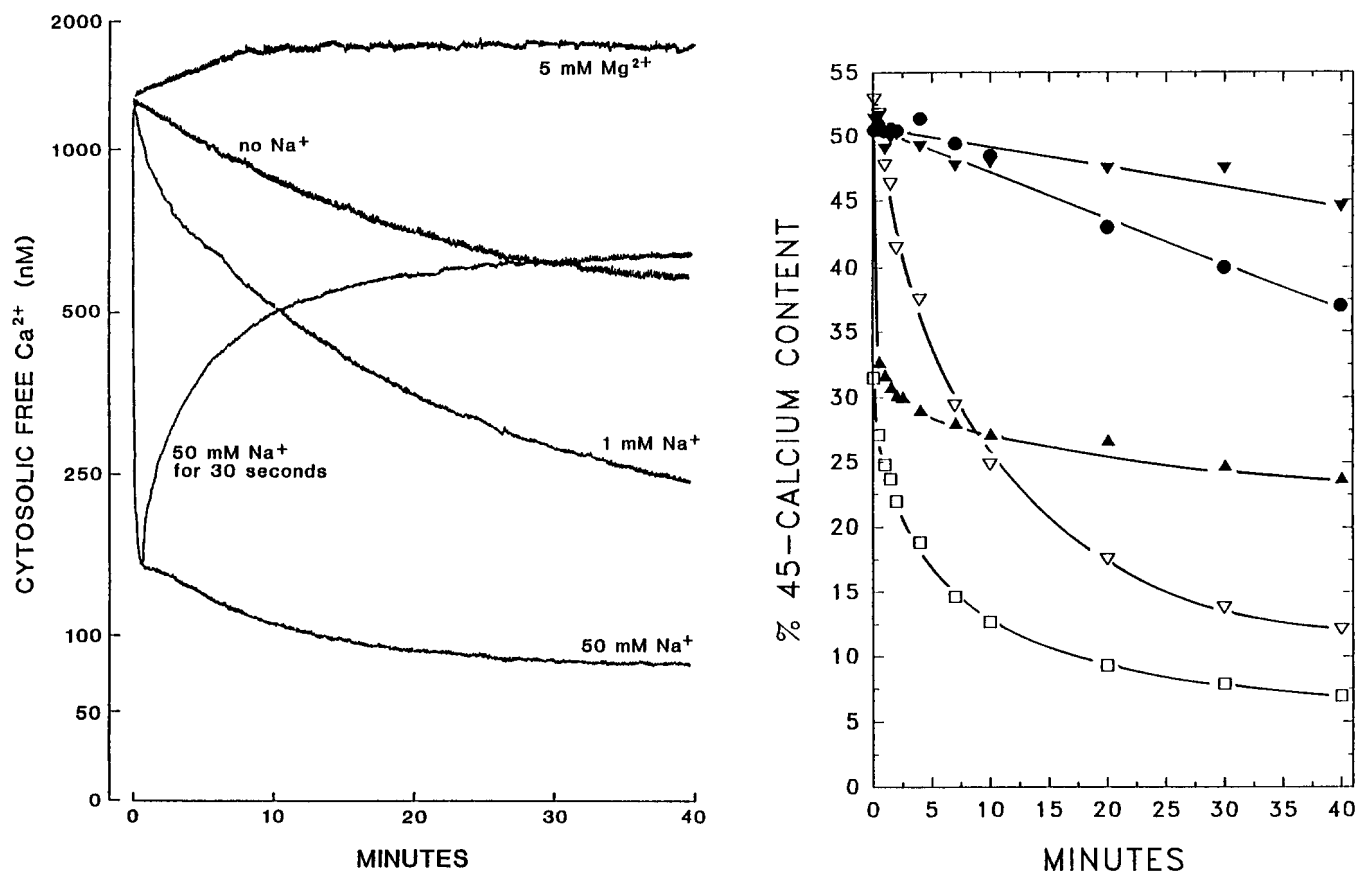


FIG. 3. Ca^{2+} release from ROS discs. Ca^{2+} - and fluo-loaded intact bovine ROS were prepared as described under Methods and were incubated in 600 mM sucrose, 20 mM Hepes (pH 7.4), 500 nM FCCP, 1 mM EGTA, and 2 mM KCl (final opsin concentration 1.5 μM). Ca^{2+} release was initiated at time 0 by addition of 50 mM NaCl (open squares), 1 mM NaCl (open triangles), 5 mM MgCl_2 (filled inverted triangles) or no addition (filled circles or labeled no Na^{+}). In one case 50 mM NaCl was added to ROS at an opsin concentration of 75 μM ; after 30 s ROS were diluted 50-fold with the above incubation medium containing 5 mM MgCl_2 (filled triangles, trace labeled 50 mM Na^{+} for 30 s). Left panel, fluo traces representing free cytosolic Ca^{2+} ; right panel, ^{45}Ca content representing total ROS Ca^{2+} . Temperature, 25 $^{\circ}\text{C}$.

observed for A23187 and would gradually decrease as the slow component of Na-Ca+K exchange removed Ca^{2+} from discs (as judged from the amplitude of A23187-induced Ca^{2+} release, e.g. Figs. 2 and 7). This was not observed, the gramicidin-induced rise in Ca^{2+} decreased much more rapidly (Fig. 6) suggesting that the rate of Ca^{2+} release from discs and its potentiation by gramicidin was not a simple function of the intradiscal Ca^{2+} content.

Comparison between A23187- and Gramicidin-induced Intracellular Ca^{2+} Release—Both A23187 and gramicidin were able to cause a rise in cytosolic free Ca^{2+} by increasing the rate of Ca^{2+} release from discs. In the case of A23187, this was due to selective permeabilization of the disc membrane by A23187 and Ca^{2+} release was most likely carried by A23187 as the release rate depended on the A23187 concentration (12). In contrast, gramicidin-induced Ca^{2+} release did not appear to be mediated by gramicidin itself as the Ca^{2+} release rate displayed little dependence on the gramicidin concentration in the range of 1 nM to 1 μM (data not illustrated). Two further differences could be noted. 1) Application of A23187 during the rapid phase of Na^{+} -stimulated Ca^{2+} release resulted in a delayed rise in cytosolic Ca^{2+} and the final Ca^{2+} level reached was less than that observed when A23187 was added during the plateau phase (Fig. 7). As pointed out before, application of A23187 before or during the fast phase of Na^{+} -stimulated Ca^{2+} release, resulted in partial or complete removal across the plasma membrane of Ca^{2+} that was released from discs by A23187. Clearly, this mechanism did not apply to gramicidin-induced Ca^{2+} release. 2) Gramicidin resulted in a persistent

rise in cytosolic free Ca^{2+} without any sign of reactivation of the exchanger. In contrast, the rise in Ca^{2+} caused by addition of A23187 resulted in partial reactivation of the exchanger and in complete removal of Ca^{2+} released from discs (12).

DISCUSSION

In darkness, a kinetic equilibrium between Ca^{2+} influx via the cGMP-gated channels and Ca^{2+} extrusion via the Na-Ca+K exchanger is thought to maintain cytosolic free Ca^{2+} at a relatively high level and an average dark value of 550 nM (range 320–880) was observed in tiger salamander rods (14). In the latter study, bright light closed all cGMP-gated channels and Ca^{2+} influx associated with it, but Ca^{2+} extrusion failed to lower free cytosolic Ca^{2+} below 50 nM, well above a value of 300 pM expected from thermodynamic considerations. Likewise, in our earlier studies on Na-Ca+K exchange in isolated bovine ROS we observed that Na^{+} -dependent Ca^{2+} extrusion did not lower free cytosolic Ca^{2+} to values below 50–100 nM, both in Ca^{2+} -containing (11) and Ca^{2+} -free media (12). In the latter study it was shown that inactivation of the Ca^{2+} extrusion mode of the exchanger accounted for these observations. The above cited studies examined lowering of cytosolic free Ca^{2+} on relatively short time scales (seconds to 10s of seconds) that are relevant for visual excitation. Inactivation of Na-Ca+K exchange may play an important role when rod photoreceptors are saturated for prolonged periods of time during daylight illumination, during which time the Na-Ca+K exchanger could lower cytosolic free Ca^{2+} toward the thermodynamic equilibrium value of <1 nM. The Na-Ca+K exchanger is thought to be

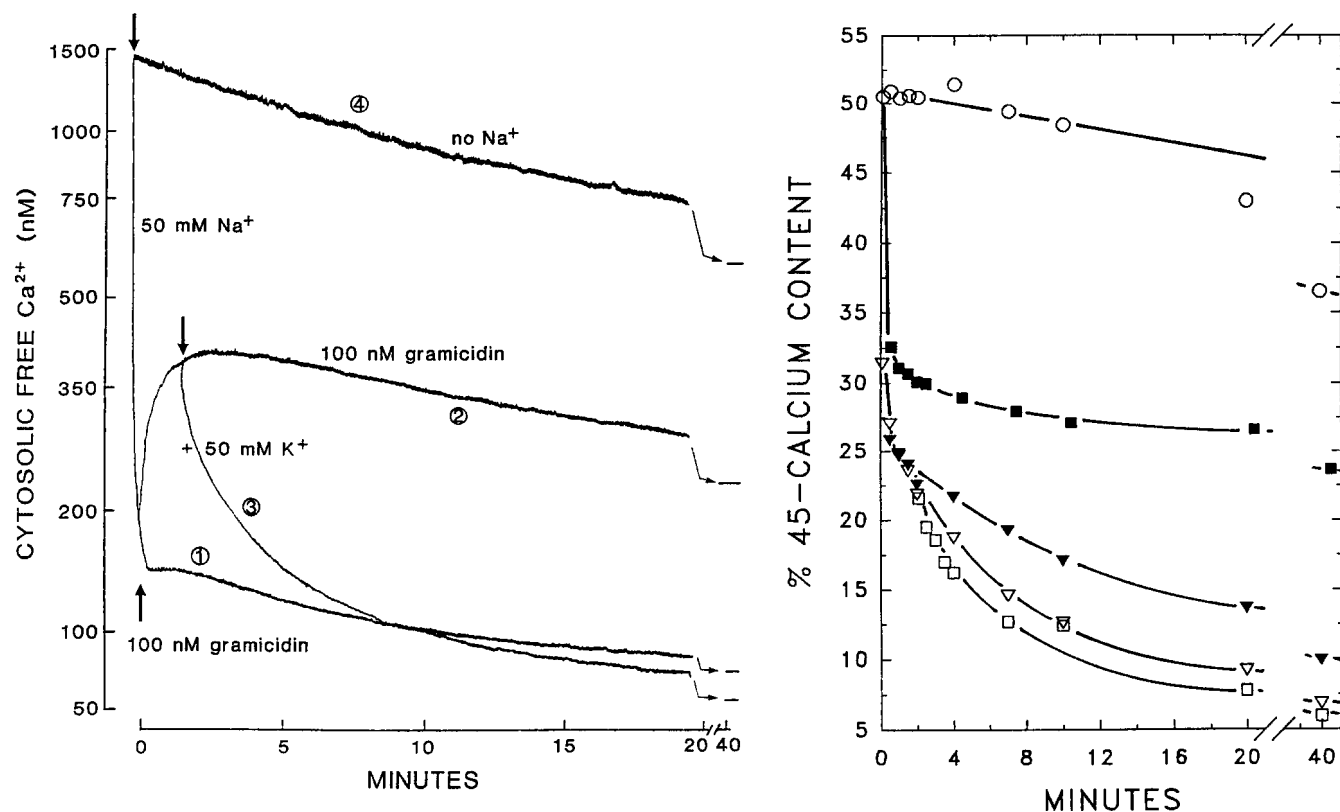


FIG. 4. Ca^{2+} release from discs and long term inactivation of Na-Ca exchange. Experimental conditions were as described in the legend of Fig. 3. Four different experimental protocols are represented and the fluo traces in the left panel are labeled accordingly: 1) 50 mM NaCl was added at time 0 (open triangles); 2) 50 mM NaCl was added at time 0, 100 nM gramicidin was added at 15 s, 50 mM KCl was added at 2 min (open squares); 3) 50 mM NaCl was added at time 0, 100 nM gramicidin was added at 15 s, 50 mM KCl was added at 2 min (open circles); 4) no addition (open circles). The ^{45}Ca panel contains the additional condition where Na^{+} -stimulated Ca^{2+} release is stopped after 30 s by 50-fold dilution in incubation medium containing 5 mM MgCl_2 (filled squares). Left panel, fluo traces representing free cytosolic Ca^{2+} ; right panel, ^{45}Ca content representing total ROS Ca^{2+} . Temperature, 25°C.

the only Ca^{2+} removal mechanism in the ROS plasma membrane, while internal Ca^{2+} sequestration and release are thought not to occur in ROS (reviewed in Refs. 9 and 17). This study had two objectives: 1) the contribution of Na-Ca+K exchange to regulation of free cytosolic Ca^{2+} was measured on time scales ranging from 1 s to 40 min, and 2) the issue of Ca^{2+} fluxes originating from internal compartments (*i.e.* ROS discs) was reexamined and their role in Ca^{2+} homeostasis in ROS considered. Our main observation is that a combination of inactivation of the Ca^{2+} extrusion mode of the Na-Ca+K exchanger coupled with Ca^{2+} release from discs is responsible for a persistent (up to 40 min) and steady cytosolic free Ca^{2+} concentration of about 100 nM, even if thermodynamic considerations would predict a free cytosolic Ca^{2+} concentration of less than 1 nM. The coupling stoichiometry of the Na-Ca+K exchanger dictates that under any reasonable physiological condition the direction of Ca^{2+} transport mediated by the exchanger is Ca^{2+} extrusion; however, the coupling stoichiometry does not appear to be involved in setting the actual free cytosolic Ca^{2+} concentration. Below I will discuss the observations that led to this conclusion, and I will discuss the factors that control the free cytosolic Ca^{2+} concentration set by fluxes via the exchanger.

Localization of Total ROS Ca^{2+} to Cytoplasm and Internal Stores—In this study, Ca^{2+} -depleted ROS were loaded with fluo-3, Ca^{2+} , and ^{45}Ca to modest total Ca^{2+} loads of 0.1–0.3 mol Ca^{2+} /mol rhodopsin (300–900 μM total Ca^{2+}). The “*in vivo*” Ca^{2+} content of either bovine or amphibian ROS is not known with certainty (reviewed in Ref. 9). In view of the potential for rapid adjustments in total internal Ca^{2+} via the Na-Ca+K

exchanger, any isolation procedure (*e.g.* cooling, altering external Na^{+} or Ca^{2+}) could conceivably result in unphysiologically low or high Ca^{2+} loads. Frog ROS with part of the inner segments attached can be purified on Percoll gradients in Ringer’s solution: they contain about 1 mol of Ca^{2+} /mol of rhodopsin and produce physiological responses to light (18, 19). Therefore, the ROS Ca^{2+} loads used in this study are not unreasonably high. The experiments illustrated in Figs. 1 and 2 demonstrate the protocol for localization of total internal ROS Ca^{2+} to the cytoplasmic and intradiscal pool, respectively: a rapid (half-time of 4–8 s) exchange of cytosolic Ca^{2+} was observed against either external Na^{+} (net Ca^{2+} efflux) or Sr^{2+} (indicative of self-exchange). A subsequent slow phase of net Ca^{2+} removal from ROS (as judged from ^{45}Ca loss) did not represent changes in cytosolic Ca^{2+} (as indicated by the steady fluo-3 fluorescence), but led to a steady reduction of intradiscal Ca^{2+} as judged by the amplitude of Ca^{2+} release from discs into the cytosol stimulated by low doses of the Ca^{2+} ionophore A23187 (Fig. 2). ^{45}Ca loaded into ROS did not to any significant amount become tightly bound to high affinity binding sites as high concentrations of A23187 caused a rapid release of all ^{45}Ca taken up (12).

Inactivation of Ca^{2+} Extrusion via Na-Ca+K Exchange Represents at Least a 20-Fold Reduction in Maximal Rate— Ca^{2+} fluxes observed in the experiments illustrated in Figs. 3 and 4 (which were done with the same ROS preparation) were calculated from the fluo experiments on the basis of a cytosolic Ca^{2+} buffer capacity of 200 μM total Ca^{2+} / μM free Ca^{2+} (12) as well as from changes in total ROS Ca^{2+} content (similar numbers were obtained in other experiments with different ROS prepa-

FIG. 5. Effect of cations on cytosolic free Ca^{2+} . Experimental conditions were as described in the legend of Fig. 3. Seven different traces are illustrated representing different cuvettes. In all cases Na^{+} -stimulated Ca^{2+} release was initiated at time 0 by addition of 50 mM NaCl (indicated by the first arrow). In one cuvette, changes in cytosolic free Ca^{2+} were followed for 4 min. To the other six cuvettes, NaCl addition was followed after 4 s by addition of 100 nM gramicidin (indicated by the second arrow), and after 30 s by addition of the indicated concentration of KCl or LiCl (indicated by the third arrow). Temperature, 25 °C.

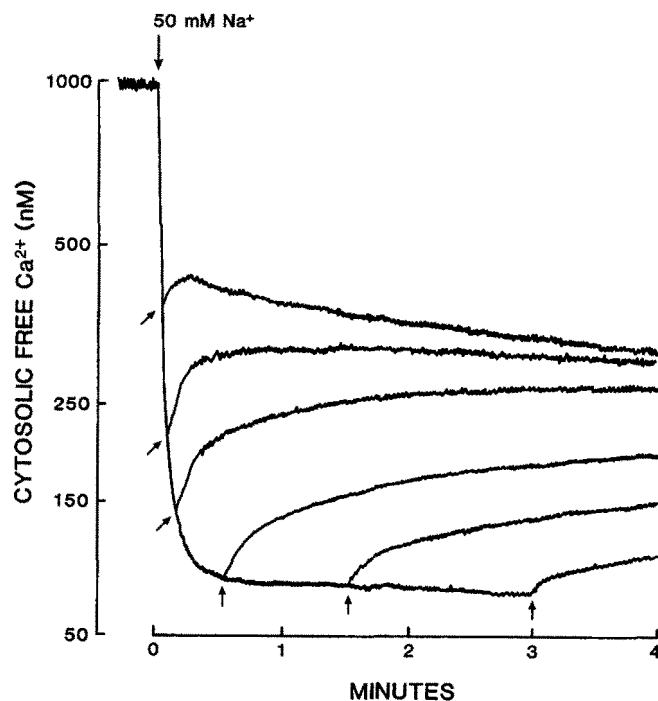
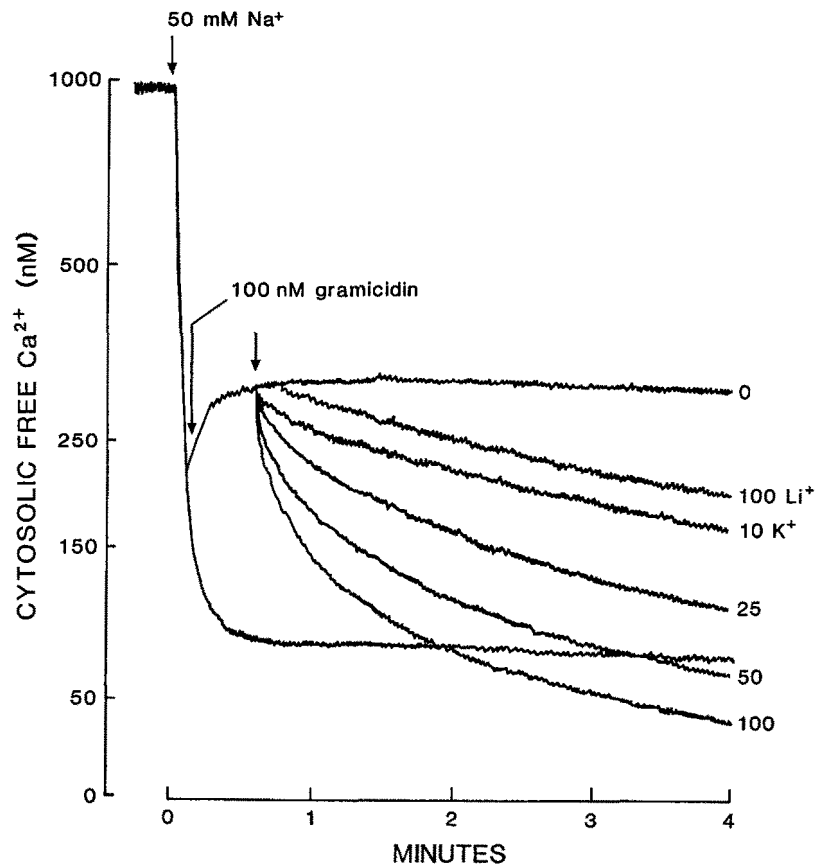


FIG. 6. Time course of gramicidin-induced Ca^{2+} release from discs. Experimental conditions were as described in the legend of Fig. 3. Seven different traces are illustrated representing different cuvettes. In all cases Na^{+} -stimulated Ca^{2+} release was initiated at time 0 by addition of 50 mM NaCl (indicated by the first arrow). In one cuvette, changes in cytosolic free Ca^{2+} were followed for 4 min. To the other six cuvettes, NaCl addition was followed after different time intervals by addition of 100 nM gramicidin (indicated by the small arrows) which resulted in a rise in cytosolic Ca^{2+} . Temperature, 25 °C.

rations). The rapid phase of Na^{+} -stimulated Ca^{2+} release caused a change in free cytosolic Ca^{2+} at an average rate over the first 5 s of 188 nM/s, equivalent to a net efflux of 3.6×10^5 cytosolic Ca^{2+} /ROS/s (fluorimetric experiment), while the total Ca^{2+} loss from ROS occurred at an efflux rate of 4.9×10^5 Ca^{2+} /ROS/s (^{45}Ca experiment). For comparison, Na-stimulated Ca^{2+} release from Ca^{2+} -enriched bovine ROS occurs at a rate of 5×10^6 Ca^{2+} /ROS/s (13, 16). After 30 s of high velocity Ca^{2+} extrusion, the total Ca^{2+} release rate dropped by a 100-fold to a rate of 4.6×10^3 Ca^{2+} /ROS/s with a negligible rate of change in cytosolic Ca^{2+} (186 Ca^{2+} /ROS/s). The average cytosolic free Ca^{2+} concentration during the first 5 s of Na^{+} -stimulated Ca^{2+} extrusion was 800 nM, while the free Ca^{2+} concentration during the first 5 min of the subsequent slow phase of Ca^{2+} extrusion was 130 nM. The apparent internal Ca^{2+} dissociation constant of the Na-Ca+K exchanger in bovine ROS is 0.9 μM (16), from which a 3.7-fold drop in Ca^{2+} extrusion rate can be calculated when the internal Ca^{2+} concentration is lowered from 800 to 130 nM (this calculation is based on a simple Michaelis-Menten occupancy of the Ca^{2+} transport site). Even if the apparent Ca^{2+} dissociation constant of the exchanger would have been 10 μM (for example due to competition with internal Na^{+}), the calculated drop in rate is only 5.8-fold. The above calculations suggest that inactivation of the Ca^{2+} extrusion mode of the exchanger reflects a drop in maximal velocity by at least 20-fold.

Ca^{2+} Release from Discs and Ca^{2+} Homeostasis in ROS—The first 5 min of the slow phase of Na^{+} -dependent Ca^{2+} extrusion illustrated in Fig. 3 occurred at a rate of 4.6×10^3 Ca^{2+} /ROS/s. Subsequently, the Ca^{2+} extrusion rate dropped to low values of 706 Ca^{2+} /ROS/s and 187 Ca^{2+} /ROS/s after 15 and 35 min, respectively. In the time interval between 30 s and 40 min after initiation of Na^{+} -dependent Ca^{2+} extrusion, free cytosolic Ca^{2+} had dropped from 139 to 83 nM, during which period net Ca^{2+} loss from the cytosol was less than 5% of the

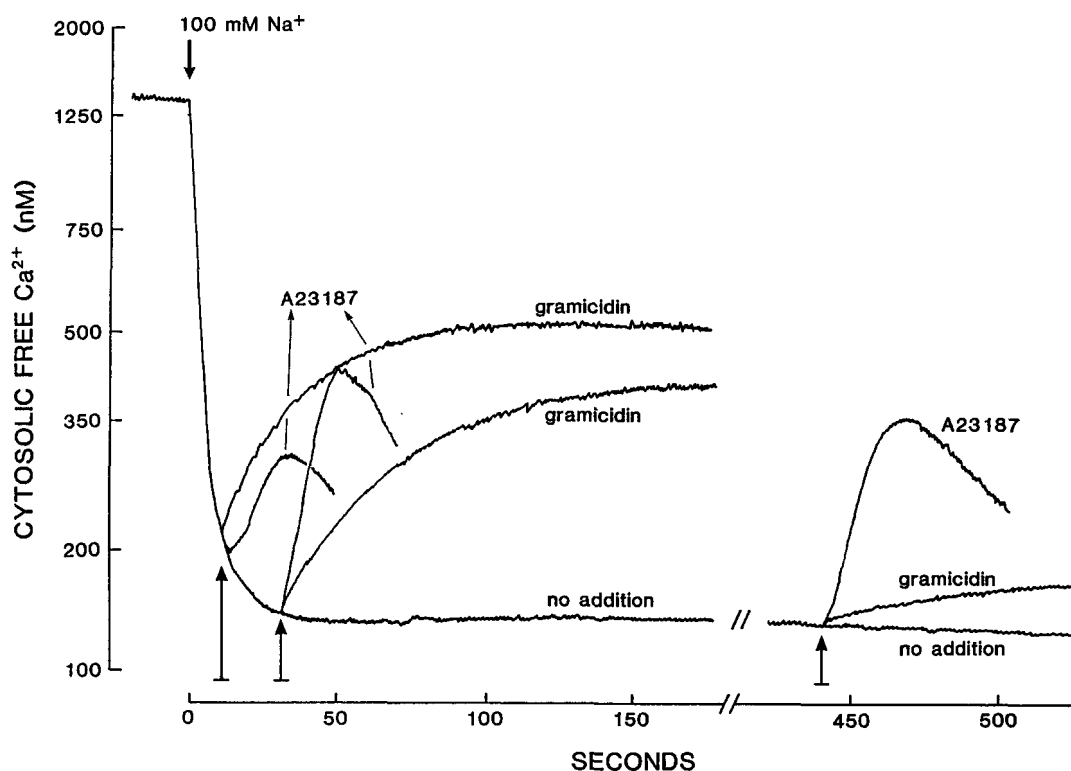


FIG. 7. **Gramicidin-induced Ca^{2+} release does not reflect the intradiscal Ca^{2+} content.** Experimental conditions were as described in the legend of Fig. 3. Seven different traces are illustrated representing different cuvettes. In all cases Na^{+} -stimulated Ca^{2+} release was initiated at time 0 by addition of 100 mM NaCl (indicated by the first arrow). In one cuvette, changes in cytosolic free Ca^{2+} were followed for 520 s. To the other six cuvettes, NaCl addition was followed after different time intervals by addition of 100 nM gramicidin or 100 nM A23187 (as indicated). Temperature, 25 °C

total Ca^{2+} loss. Ca^{2+} release from discs was responsible for the steady cytosolic Ca^{2+} concentration and the initial release rate from discs was 2–4 nM/s or $3.8\text{--}7.6 \times 10^3 \text{ Ca}^{2+}/\text{ROS/s}$ (Fig. 3), very similar to the rate calculated above for the first 5 min of the slow Ca^{2+} release phase. As cytosolic free Ca^{2+} did not change very much over the next 35 min, the dramatic drop in total Ca^{2+} release rate from ROS must have reflected an equally large drop in Ca^{2+} release rate from discs. We previously reported that inactivation of the Ca^{2+} extrusion mode of the Na-Ca+K exchanger was abolished when intradiscal Ca^{2+} was removed by selective Ca^{2+} permeabilization of the disc membranes with low doses of A23187 and we proposed a role for intradiscal Ca^{2+} in inactivation (12). This suggestion was based on the absence of a long lived steady cytosolic free Ca^{2+} concentration after pretreatment with A23187 and subsequent Na^{+} -dependent Ca^{2+} -extrusion. Either intradiscal Ca^{2+} is somehow directly involved in inactivation or an equilibrium between Ca^{2+} release from discs and inactivated Na^{+} -dependent Ca^{2+} extrusion determines the steady state cytosolic free Ca^{2+} concentration and removal of intradiscal Ca^{2+} by A23187 pretreatment disturbs this equilibrium. Such a scheme would also explain the observation that the free Ca^{2+} concentration at which the exchanger inactivated increased when the intradiscal Ca^{2+} content was increased (not illustrated; see also Fig. 8 in Ref. 11).

Does Cytosolic Free Ca^{2+} Control Inactivation of Ca^{2+} Extrusion via Na-Ca+K Exchange?—Above it is suggested that the long lived steady cytosolic free Ca^{2+} concentration reflects an apparent equilibrium between Ca^{2+} release from discs and its subsequent removal across the plasma membrane by an inactivated Na-Ca+K exchanger. However, the low velocity Ca^{2+} extrusion mode operated over a more than 20-fold range (from $4.6 \times 10^3 \text{ Ca}^{2+}/\text{ROS/s}$ initially to a final value of $187 \text{ Ca}^{2+}/\text{ROS/s}$), while free cytosolic Ca^{2+} changed very little from 139

nM to 83 nM (Fig. 3). This suggests an alternative interpretation in which a sensor controls cytosolic free Ca^{2+} either by adjusting the Ca^{2+} extrusion rate or the Ca^{2+} release rate from discs. Perhaps a Ca^{2+} sensor mediates inactivation of the Ca^{2+} efflux mode when the appropriate free cytosolic Ca^{2+} concentration, far above that expected from the coupling stoichiometry, is reached. Two properties of this putative sensor can be noted. 1) It must cause prolonged inactivation of the Ca^{2+} extrusion mode. When cytosolic free Ca^{2+} was raised to 500 nM by A23187-induced Ca^{2+} release from discs, it took seconds for the Ca^{2+} extrusion rate to reactivate and only a partial reactivation was observed (12) (Fig. 7). 2) The Ca^{2+} release rate inactivated within a very narrow range of free Ca^{2+} concentration (e.g. Fig. 1), which suggests a highly cooperative mechanism. The cytosolic domain of the Na-Ca+K exchanger contains multiple repeats of the highly acidic decamer AQIEGEDDQ in which the underlined residues are conserved (20). These repeats may represent allosteric Ca^{2+} binding sites that are involved in the inactivation process. Inactivation of Na-Ca+K exchange is limited to the Ca^{2+} extrusion mode and no internal Ca^{2+} dependence of Ca^{2+} influx via reverse exchange is apparent (10).

At least two factors other than Ca^{2+} appear to influence the putative intracellular Ca^{2+} sensor, intracellular Na^{+} and intracellular K^{+} (Fig. 4). Addition of the alkali cation ionophore gramicidin in the presence of external Na^{+} caused a persistent increase in cytosolic free Ca^{2+} suggesting a shift in the Ca^{2+} sensor controlling inactivation to a higher free Ca^{2+} concentration (Figs. 4–6). Gramicidin equilibrates internal and external alkali cation concentrations when applied to bovine ROS (16, 21) and this suggests that an increase in intracellular Na^{+} caused the above-noted shift in the Ca^{2+} sensor. K^{+} reversed the action of Na^{+} and resulted in lower values of cytosolic free Ca^{2+} (Figs. 4–6). The gramicidin-induced and Na^{+} -dependent

rise in cytosolic Ca^{2+} is useful to analyze the mechanisms involved in regulating cytosolic free Ca^{2+} . First of all, the Na-Ca+K exchanger "allowed" the free cytosolic Ca^{2+} to increase to a new set value despite the fact that it still mediated net Ca^{2+} efflux from ROS (Fig. 4). This is perhaps the most compelling evidence that the exchanger possesses a sensor to set free cytosolic Ca^{2+} at levels far above those predicted from the coupling stoichiometry. In our experiments no Ca^{2+} was present in the external medium and, therefore, the exchanger can only dictate free cytosolic Ca^{2+} by regulating its Ca^{2+} extrusion rate. Moreover, exchanger-mediated Ca^{2+} influx to raise cytosolic Ca^{2+} would violate the thermodynamic consequences of the coupling stoichiometry. Ca^{2+} release from discs was required as a source of Ca^{2+} to raise cytosolic Ca^{2+} and this suggests that Ca^{2+} efflux from discs is important in keeping cytosolic free Ca^{2+} well above values predicted from the coupling stoichiometry. Further experiments are under way to delineate further 1) the role of Ca^{2+} release from discs in Ca^{2+} homeostasis in ROS, 2) the mechanism(s) responsible for Ca^{2+} uptake in and Ca^{2+} release from discs, and 3) the secondary cation binding sites that are proposed to constitute a sensor that controls the Ca^{2+} extrusion rate of the Na-Ca+K exchanger.

Acknowledgment—I gratefully acknowledge the expert technical assistance of Robert T. Szerencsei.

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